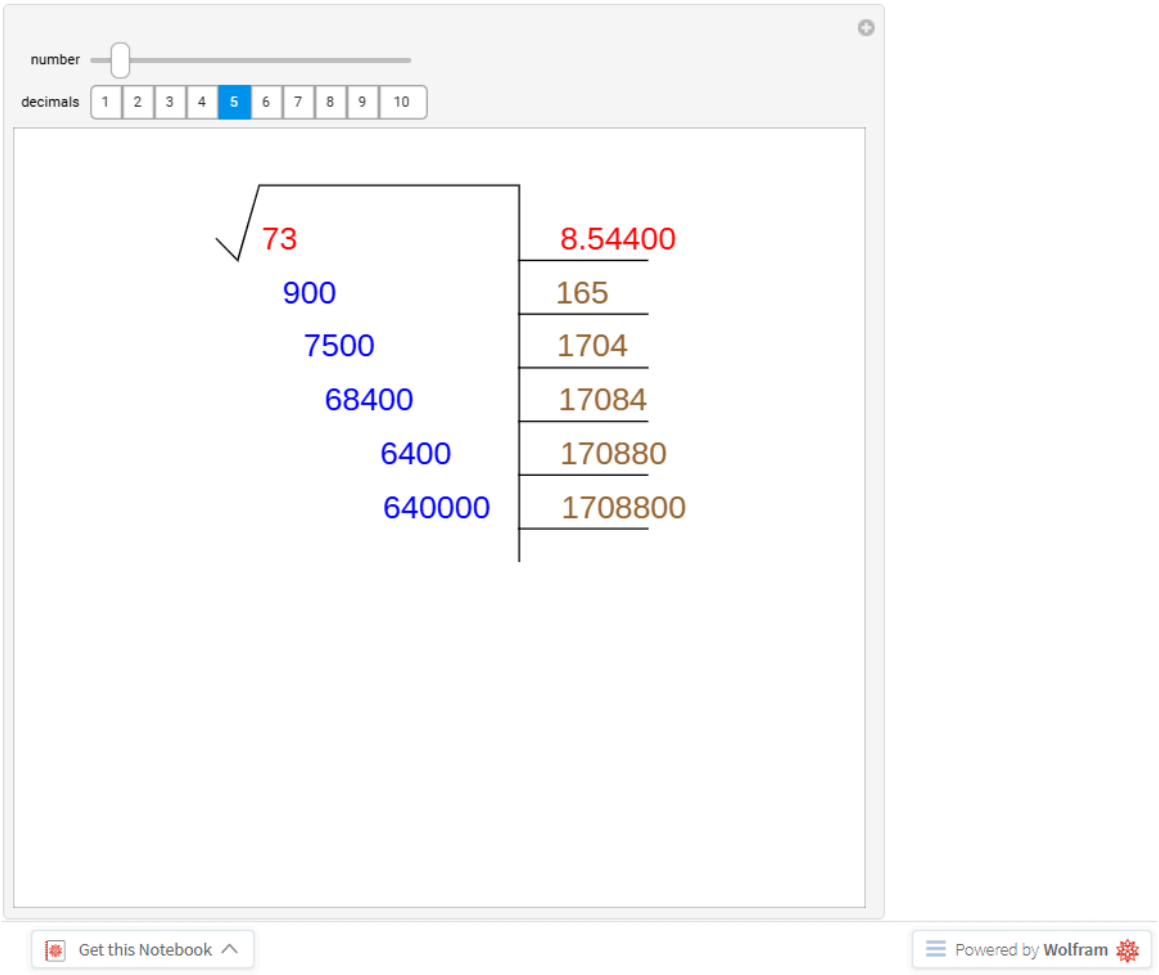
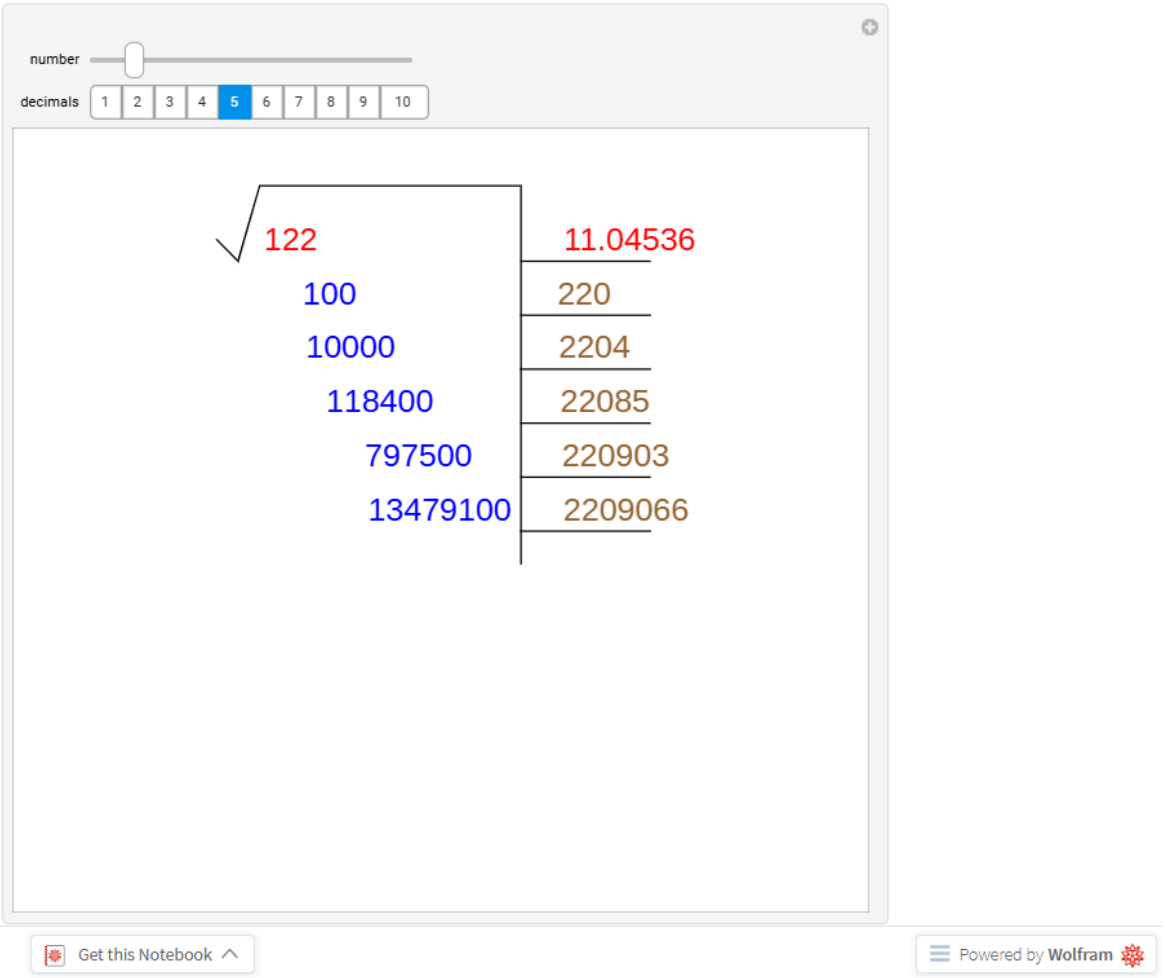
NAME: MOHAMMED MUIN
CLASS: D7B
ROLL NO: 45

An Algorithm to Compute Square Roots of Integers

This Demonstration shows a method of computing the square root of a positive number, in this case a nonsquare integer less than 1000. This method was very popular and taught in schools before the advent of calculators but still retains its usefulness in understanding and practicing algorithmic manipulation.





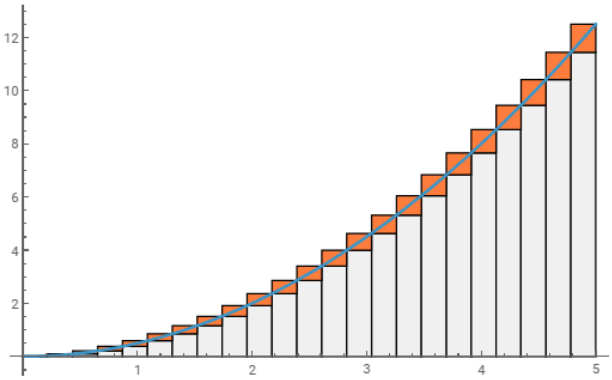
Continuous Functions Are Integrable

This Demonstration illustrates a theorem from calculus: A continuous function on a closed interval [*a*,*b*] is integrable, which means that the difference between the upper and lower sums approaches 0 as the length of the subintervals approaches 0.

number of rectangles 23

function x/2 | $x^2/2$ | $x^3/4$ | log(x + 1) | sin(x/5)

upper – lower integral sum = 2.7174
actual area = 20.8333



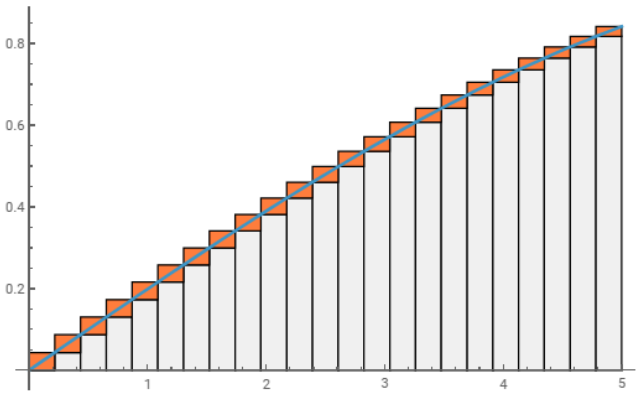
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number of rectangles 23

function x/2 | $x^2/2$ | $x^3/4$ | log(x + 1) | sin(x/5)

upper – lower integral sum = 0.1829
actual area = 2.2985

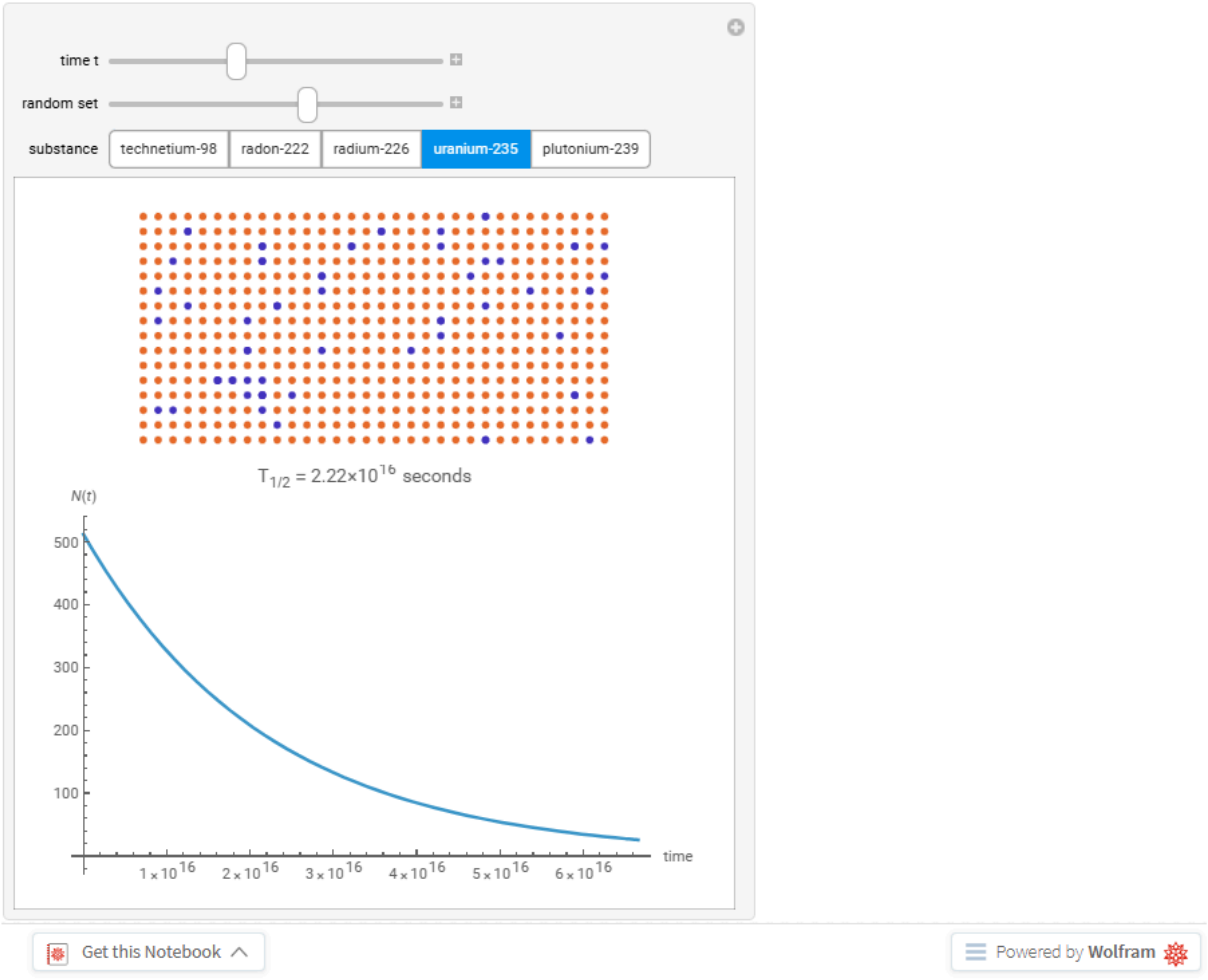


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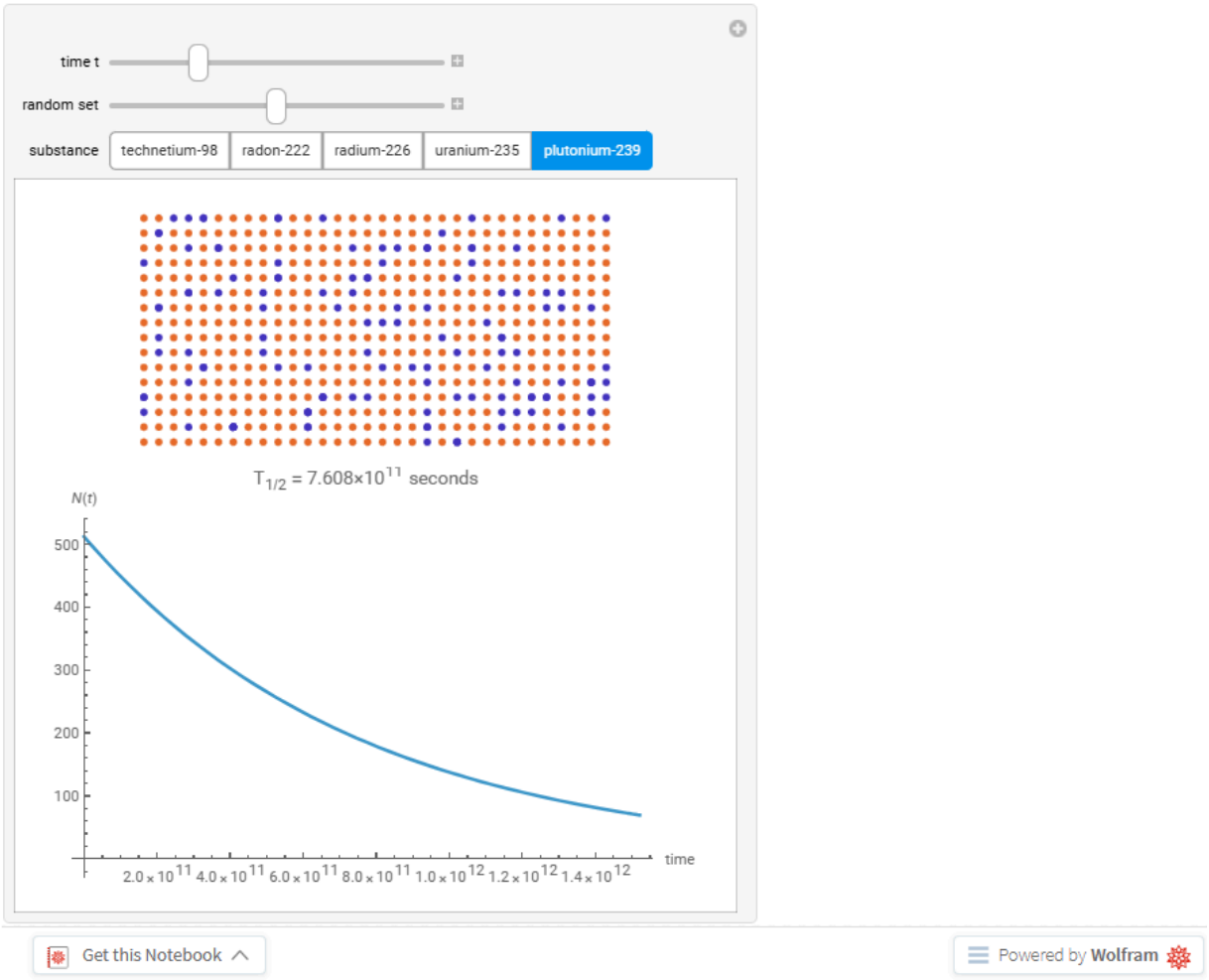
Half-Life of a Radioactive Element

The half life of a radioactive element is the time needed for half of the material to decay. The blue and orange points represent the original number of radioactive nuclei and those that decay; the number of blue points decreases by half at each step in time. The plot shows the population N(t) of radioactive nuclei at anytime(in T1/2 units).



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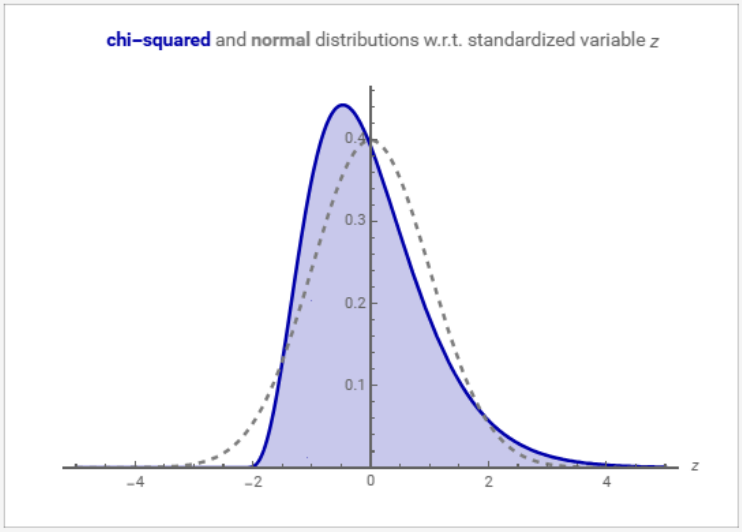
Chi-Squared Distribution and the Central Limit Theorem

This Demonstration explores the chi-squared distribution for large degrees of freedom v, which, when suitably standardized, approaches a standard normal distribution as v∞ by the central limit theorem. In this Demonstration, v can be varied between 1 and 2000 and either the PDF or CDF of the chi-squared and standard normal distribution can be viewed. You can also see the difference between the two distributions, which becomes useful for large v.

degrees of freedom ν 9

distribution function pdf cdf

show difference



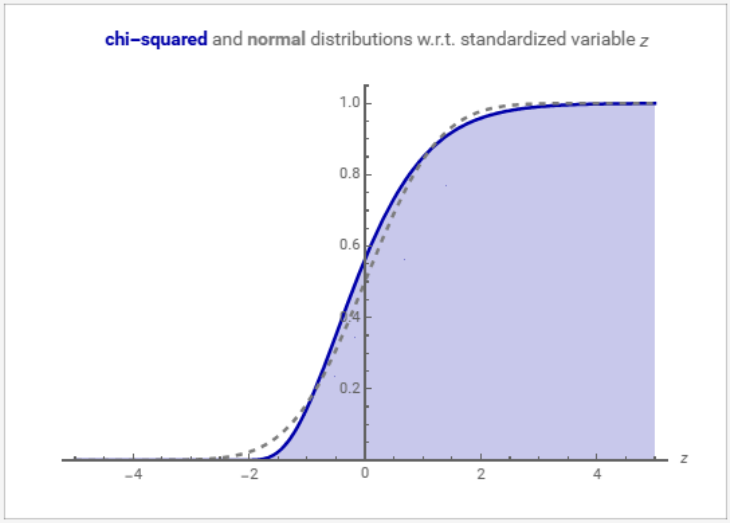
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distribution function pdf cdf

show difference



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